

Martel Vasquez

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[Linkedin](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

Bachelor of Science in Computer Science
George Mason University
GPA: 3.69

Expected graduation: August 2025
Fairfax, VA

Dean's List: Spring 2024, Summer 2024, Spring 2025

TECHNICAL PROJECTS:

Java

- Tetris Game: Built a Tetris game using 2D arrays, improving performance by **10%** with redraw logic.
- Cache Simulator: Implemented FIFO, LFU, and LRU cache strategies with Linked Lists and HashMap, reducing retrieval times by **20%**.
- Image Processor: Utilized a Quadtree data structure to process PGB images, improving processing speed by **25%**.

HTML, CSS & JavaScript

- Portfolio: Designed and built a responsive website to showcase projects, skills, and technical achievements, enhancing web development proficiency and user experience.

Dart/Flutter

- Flextimer App: Developed a workout timer app that allows users to save workouts, target muscle groups, and receive exercise recommendations. Contributed to requirements, advanced calculator features, and UI design (light/dark mode, app info), applying software engineering principles and teamwork.

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- OS/161 Synchronization & Traffic Management: Developed custom locks for mutual exclusion and designed a traffic system for a three-way stop. Used `thread_sleep()` and `thread_wakeup()` to manage threads, prioritizing cars and avoiding deadlocks, improving system efficiency by **30%**.
- Oturn CPU Scheduler: Engineered a CPU scheduler for the StrawHat Virtual Machine using three linked list queues, enhancing efficiency by **15%**.

Python

- Pacman Maze Search: Engineered search algorithms (DFS, BFS, UCS, and A*) for a Pacman maze agent. Designed custom heuristics to efficiently solve fixed food, corner, and food collection challenges.

EXPERIENCE:

Footwear Associate

Dicks Sporting Goods

August 2021 – Present
Fairfax, VA

- Drove a **35%** sales increase through enhanced customer interaction and product knowledge.
- Elevated store's customer satisfaction by **20%**, earning praise for professionalism and team support.

SKILLS:

- **Languages:** Java (Advanced), Python, C, HTML, CSS, JavaScript, SQL, Dart
- **Tools/Technologies:** GitHub, GitLab, Linux, VS Code, IntelliJ, CLion, JUnit

RELEVANT COURSEWORK:

- Data Structures, Software Engineering, Artificial Intelligence, Object-Oriented Programming, Formal Methods & Models, Computer Systems & Programming, Operating Systems, Analysis of Algorithms, Secure Programming and Systems, Database Concepts, Comparative Programming Languages